ModelIt!

Student Activity Pack

Why Can't Humans Photosynthesize?

Modeling Energy Flow Between Photosynthesis and Cellular Respiration

Grade 10L1 | NGSS HS-LS1-7, HS-LS2-5 | Energy and Matter

STUDENT NAME _____ DATE _____

ModelIt! — Discovery Collective

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Table of Contents

ModelIt!

1 Pre-Assessment 3

2 Scoring Rubric 4

3 Component Analysis 5

4 Model Recording Page 6

5 Simulation Observations 7

6 Research & Extend 8

7 Reflection Questions 9

8 STEM Challenge 10

1 Pre-Assessment

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What Do You Know? Answer each question below to the best of your ability. There are no wrong answers — this helps your teacher understand what you already know!

1 Why do plants need sunlight to grow but humans don't?

2 What do you think would happen if human skin contained chlorophyll — could we skip eating?

3 Draw a diagram showing how energy flows from the sun through a plant and eventually to a human.

4 What is the relationship between the food you eat and the energy your cells use?

2 Scoring Rubric

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Your teacher will use this rubric to assess your work. Read through each level to understand what is expected!

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

3 Component Analysis

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What is a Component?

A **component** is a part of a system that plays a specific role. Components can be **EXTERNAL** (come from outside the system) or **INTERNAL** (happen within the system).

Sort These Components

Sunlight Intensity | Chlorophyll Concentration | Glucose Production Rate | Cellular Respiration Rate | Net Energy Output

EXTERNAL	INTERNAL

Explain your reasoning: Why did you sort each component the way you did?

4 Model Recording Page

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Draw your model showing all components and their relationships. Use the relationship key below to label your arrows!

+ **POSITIVE:** When A increases, B also increases

− **NEGATIVE:** When A increases, B decreases

Draw your model here

? Explain ONE relationship in your model. What happens when one component changes?

5 Simulation Observations

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Run each scenario in ModelIt! First, write your **prediction**. Then run the simulation and record what you **observe**.

1 Scenario 1: Optimal Plant

Sunlight: Maximum | Chlorophyll: High | Plant scale

MY PREDICTION

WHAT I OBSERVED

2 Scenario 2: Photosynthetic Human

Surface area: 1.7m^2 | Respiration: 2,000 kcal/day

MY PREDICTION

WHAT I OBSERVED

3 Scenario 3: Scaling Problem

Organisms from bacteria to human scale

MY PREDICTION

WHAT I OBSERVED

6 Research & Extend

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> Real systems are more complex than our model! Here are additional components scientists study. Choose one and explain how it could connect to our model.

Possible New Components

CO_2 Concentration — The amount of carbon dioxide available for the Calvin cycle reactions — increasing CO_2 up to a saturation point increases glucose production, but beyond that point, other factors become limiting

Leaf Area Index — The total photosynthetic surface area relative to ground area — forests have LAI of 6-8 (many overlapping leaf layers), while a flat human has an LAI of about 0.4, severely limiting light capture

Metabolic Efficiency — The percentage of glucose energy that is successfully converted to ATP during cellular respiration — approximately 38-40% in aerobic conditions, with the remaining 60% lost as heat

My Extension

A Which new component would you add to the model?

B How would it connect to the existing components? (Draw or describe)

C What evidence or reasoning supports your connection?

7 Reflection Questions

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The Big Question

Plants convert sunlight into food using nothing but water and CO_2. Why can't humans do the same — and what would happen if we could? Use evidence from your simulation to support your answers.

1. **How does your model demonstrate why large animals cannot survive on photosynthesis alone?**

2. **What does the surface-area-to-volume ratio teach us about why plants and animals evolved such different body plans?**

3. **How are photosynthesis and cellular respiration complementary reactions that drive the global carbon cycle?**

4. **If a biotech company engineered human skin with chlorophyll, would it solve world hunger? Use your model to evaluate.**

5. **What are the limitations of modeling the energy balance with only five components?**

8 STEM Challenge: Design a Bio-Solar Energy Harvesting System

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YOUR ENGINEERING MISSION

Design a biological or bio-inspired system that captures solar energy more efficiently than natural photosynthesis and converts it to usable energy for human applications.

THE SCENARIO

A biotech startup is exploring bio-solar energy — using biological or bio-inspired systems to capture sunlight more efficiently than traditional solar panels. Natural photosynthesis is only 1-2% efficient, while silicon solar cells reach 20-25%. Your team must design a system that bridges this gap — either by improving biological photosynthesis through bioengineering or by creating bio-inspired artificial systems.

Engineering Design Thinking

1. WILL YOUR SYSTEM USE LIVING ORGANISMS, BIO-INSPIRED MATERIALS, OR A HYBRID APPROACH?
2. WHAT SPECIFIC BOTTLENECK IN PHOTOSYNTHESIS DOES YOUR DESIGN ADDRESS?
3. HOW WILL YOU CONVERT THE CAPTURED ENERGY INTO A FORM USABLE FOR HUMAN APPLICATIONS?
4. WHAT ARE THE SCALABILITY AND ENVIRONMENTAL IMPLICATIONS OF YOUR DESIGN?

My Design

DESIGN AREA — DRAW YOUR SOLUTION

N

! Explain your design using scientific evidence from the simulation:

Real Career Connection

Bioenergy Researchers develop biological and bio-inspired systems for energy capture and conversion, including artificial photosynthesis, algal biofuels, and bio-solar cells. They work at universities, national laboratories, and biotech companies, earning $65,000–$130,000/year. Metabolic Engineers who redesign biological pathways for energy and chemical production earn $80,000–$150,000/year.